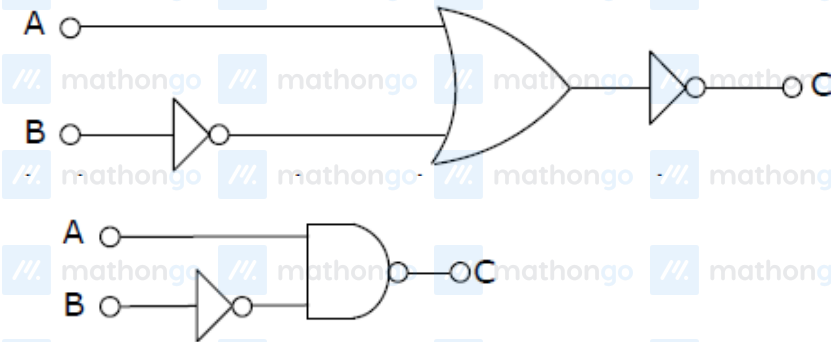


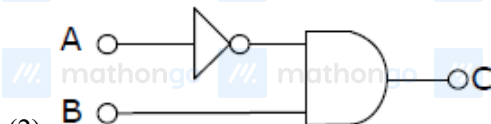
Q1: 24 Feb (Shift 1) - Numerical

An audio signal $v_m = 20 \sin 2\pi(1500t)$ amplitude modulates a carrier $v_c = 80 \sin 2\pi(100,000t)$. The value of percent modulation is

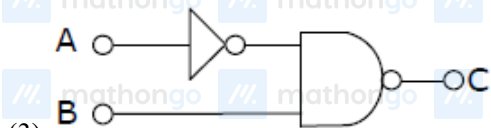
Q2: 24 Feb (Shift 2) - Single Correct



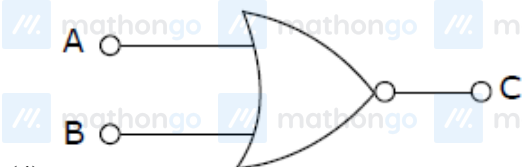
(1)



(2)



(3)



(4)

Q3: 24 Feb (Shift 2) - Numerical

A signal of 0.1 kW is transmitted in a cable. The attenuation of cable is -5 dB per km and cable length is 20 km. the power received at receiver is 10^{-x} W. The value of X is ____ [Gain in dB $= 10 \log_{10} \left(\frac{P_0}{P_i} \right)$]

Q4: 25 Feb (Shift 1) - Single Correct

Given below are two statements :

Statement I : A speech signal of 2kHz is used to modulate a carrier signal of 1MHz. The bandwidth requirement for the signal is 4kHz.

Statement II : The side band frequencies are 1002kHz and 998kHz. In the light of the above statements, choose the correct answer from the options given below :

- (1) Both statement I and statement II are false
- (2) Statement I is false but statement II is true
- (3) Statement I is true but statement II is false
- (4) Both statement I and statement II are true

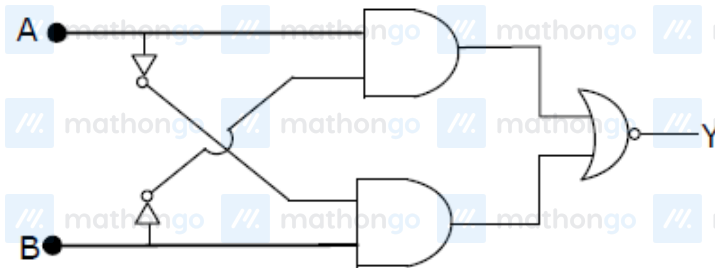
Q5: 25 Feb (Shift 2) - Single Correct

if a message signal of frequency ' f_m ' is amplitude modulated with a carrier signal of frequency f_c and radiated through an antenna, the wavelength of the corresponding signal in air is:

- (1) $\frac{c}{f_c + f_m}$
- (2) $\frac{c}{f_c - f_m}$
- (3) $\frac{c}{f_m}$
- (4) $\frac{c}{f_c}$

Q6: 25 Feb (Shift 2) - Single Correct

The truth table for the following logic circuit is:



A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(1)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(3)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

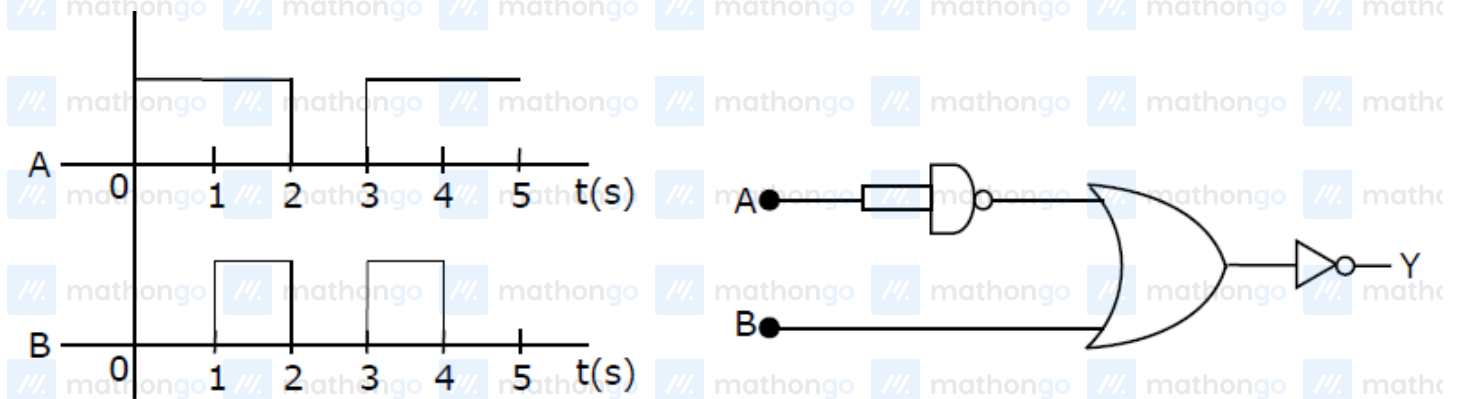
(4)

Q7: 26 Feb (Shift 1) - Numerical

The maximum and minimum amplitude of an amplitude modulated wave is 16 V and 8 V respectively. The modulation index for this amplitude modulated wave is $x \times 10^{-2}$. The value of x is

Q8: 26 Feb (Shift 2) - Single Correct

Draw the output Y in the given combination of gates.



**Q9: 26 Feb (Shift 2) - Numerical**

If the highest frequency modulating a carrier is 5 kHz, then the number of AM broadcast stations accommodated in a 90 kHz bandwidth are

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathc

Answer Key

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathc

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathc

Q1 (25)

Q2 (2)

Q3 (8)

Q4 (4)

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathc

Q5 (4)

Q6 (4)

Q7 (33)

Q8 (1)

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathc

Q9 (9)

// mathongo // mathongo // mathongo // mathongo // mathongo // mathongo // mathc

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Hints and Solutions

MathonGo

Q1 (25)

Given : $v_m = 20 \sin \left[100\pi t + \frac{\pi}{4} \right]$

$v_c = 80 \sin \left[10^4 \pi t + \frac{\pi}{6} \right]$

we know that, modulation index $= \frac{A_m}{A_c}$

from given equations, $A_m = 20$ and $A_c = 80$

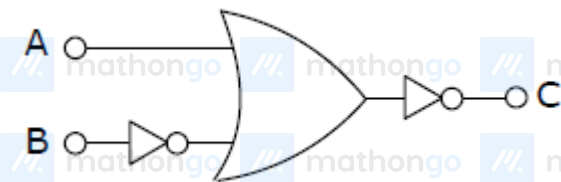
percentage modulation index $= \frac{A_m}{A_c} \times 100$

$\Rightarrow \frac{20}{80} \times 100 = 25\%$

The value of percentage modulation index is

$= 25$

Q2 (2)



$C = \overline{\bar{A} + \bar{B}}$

$C = \bar{A} \cdot B$

Q3 (8)

Power of signal transmitted : $P_i = 0.1 \text{ KW} = 100 \text{ w}$

Rate of attenuation $= -5 \text{ dB/Km}$

Total length of path $= 20 \text{ km}$

Total loss suffered $= -5 \times 20 = -100 \text{ dB}$

Gain in dB $= 10 \log_{10} \frac{P_0}{P_i}$

$-100 = 10 \log_{10} \frac{P_0}{P_i}$

$\Rightarrow \log_{10} \frac{P_0}{P_i} = 10$

$\Rightarrow \log_{10} \frac{P_i}{P_0} = \log_{10} 10^{-10}$

$\Rightarrow \frac{100}{P_0} = 10^{-10}$

Hints and Solutions

MathonGo

$$\Rightarrow P_0 = \frac{1}{10^8} = 10^{-8}$$

$$\therefore x = 8$$

Q4 (4)

$$\text{Side band} = (f_c - f_m) \text{ to } (f_c + f_m) = (1000 - 2)\text{KHz to } (1000 + 2)\text{KHz} \\ = 998\text{KHz to } 1002\text{kHz}$$

$$\text{Band width} = 2f_m$$

$$= 2 \times 2\text{KHz}$$

$$= 4\text{KHz}$$

Both statements are true

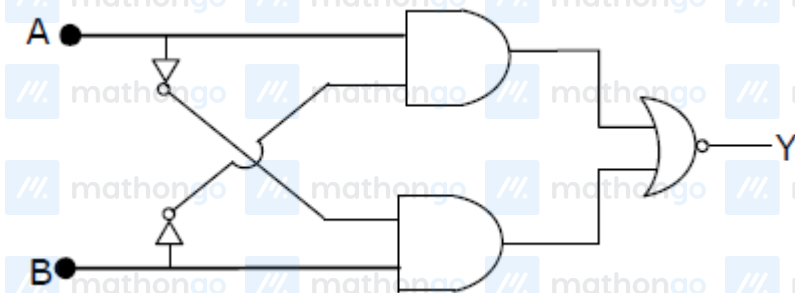
Q5 (4)

Given frequency of message signal = f_m

frequency of carrier signal = f_c

the wavelength of the corresponding signal in air is $\Rightarrow \lambda = \frac{c}{f_c}$

Q6 (4)



If $A = B = 0$ then output $y = 1$

If $A = B = 1$ then output $y = 1$

Q7 (33)

Hints and Solutions

MathonGo

$$A_m = \frac{A_{\max} - A_{\min}}{2}$$

$$A_C = \frac{A_{\max} + A_{\min}}{2} \left[\begin{array}{l} A_{\max} = 16 \text{ V} \\ A_{\min} = 8 \text{ V} \end{array} \right]$$

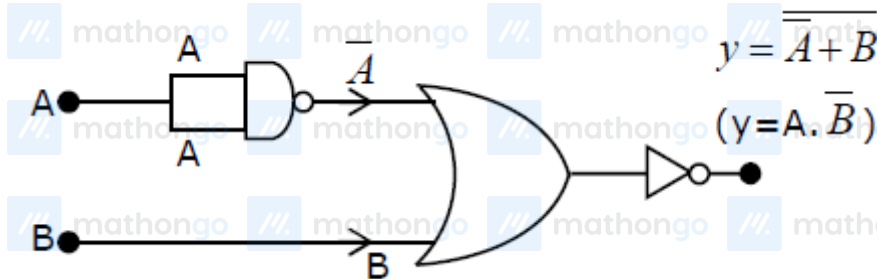
$$\text{Modulation index (mi)} = \frac{A_m}{A_C} = \frac{\frac{A_{\max} - A_{\min}}{2}}{\frac{A_{\max} + A_{\min}}{2}} = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

$$mi = \frac{16-8}{16+8} = \frac{8}{24} = \frac{1}{3} = 0.33$$

$$mi = 33 \times 10^{-2}$$

$$X = 33$$

Q8 (1)



Find output expression $y = A \cdot B$

Inputs

A	B	$y = A \cdot B$
1	0	1
1	1	0
0	0	0
1	1	0
1	0	1

Q9 (9)

$$\text{No. of station} = \frac{\text{Band width}}{2 \times \text{Highest Band width}}$$

$$\Rightarrow \frac{90}{2 \times 5}$$

$$\Rightarrow 9$$

